Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	***Provably Safe Autonomous Coordination for Congested Space and Air Operations***				
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (x), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )**				
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**			**Technology Field 2 (**Large Category**)**	
	Mechanical Materials			***Mechanical Materials***	
	Medium Category		Small Category	Medium Category	Small Category
	Aerospace Systems		***Aerospace System Software***	***Robotics/Automation Machinery***	***Robot Control and Intelligence Technologies***
Concept of Proposed Technology	*This project develops safe and scalable methodologies for integrating autonomous vehicles into existing space and air traffic (STM/ATM) systems. Aligned with Korea's growing low Earth orbit (LEO) satellite ecosystem and urban air mobility (UAM) initiatives, the proposed reseach provides a theoretically grounded system-level framework that addresses fundamental challenges to autonomy adaptatxion in congested aerial and orbital environments.* *Current ATM/STM operational practices rely heavily on human-in-the-loop, heuristic decision making for high-level guidance. While currently effective, this paradigm does not scale as these environments become increasingly congested with vehicles that may differ in objectives, means of communication,*				

	and maneuver capabilities. These limitations are especially pronounced in multi-operator settings, where coordination and information sharing are inherently constrained. *The proposed technology introduces a conflict resolution architecture that integrates* *(i) formal safety allocation,* *(ii) early maneuver guidance under large state uncertainty,* *(iii) intent-aware prediction to enable proactive traffic management in both orbital and airspace domains.* *Furthermore, the proposed research targets key bottleneck questions that cannot be resolved through empirical tuning alone:* *1) How existing guarantees scale from local conflict resolution to* ***system-level guarantees****?* *2) What is the appropriate framework for* ***early conflict resolution****, where close-range maneuvers are dangerous and resource-inefficient, and mid-range conflict resolution may create conflict chains among more vehicles?* *3) How can satellite constellations be safely managed under kilometer-scale uncertainty, complex orbital dynamics, and interactions with uncontrolled resident space objects (RSOs), while maintaining operational performance?*
Rationale for Joint R&D Project	*Korea's rapidly expanding LEO satellite operations and urban air mobility (UAM) initiatives create an urgent need for scalable, autonomous traffic management technologies that go beyond human-centered and heuristic solutions. Korean industries bring strong capabilities in platform development, systems integration, and operational deployment, while this project contributes theoretically grounded decision and safety frameworks required for multi-operator STM/ATM at scale. A joint R&D effort enables early validation of these autonomy technologies on Korean platforms and operational scenarios, accelerating transition from research to deployable products. This collaboration positions Korean industry at the forefront of next-generation autonomous traffic management for both space and air domains, supporting national competitiveness and long-term commercialization.*
Key Words	1. *Risk-driven satellite guidance* 2. *Multi-agent autonomous systems* 3. *Decentralized coordination* 4. *Scalable autonomy architectures* 5. *Urban air mobility.*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		**Start of the Project**	Level ( )	**End of the Project**	Level ( )
R&D Objectives		This R&D delivers practical autonomy technologies that enable safe, scalable management of dense space and air traffic. The research supports real-time decision support and automated maneuver coordination for heterogeneous vehicles, reducing operational burden, improving safety and efficiency, and accelerating deployment of autonomous traffic management services for commercial satellite and air mobility operators.			
R&D Target Specifications		*The criteria to measure success in the proposed research include:* • *System-Level Safety Certification Framework: Develop formal, scalable safety certificates that translate global STM/ATM safety requirements into deployable rules, enabling industry to certify autonomous operations across fleets rather than vehicle-by-vehicle.* • *Corrective Maneuver Allocation Algorithms: Create theoretically grounded algorithms that determine which vehicles should maneuver and how much under limited resources, enabling efficient, automated conflict resolution without manual operator intervention.* • *Privacy-Preserving Multi-Operator Coordination: Coordination methods that ensure safety with information-efficient sharing between operators.* • *Scalable STM/ATM Middleware Prototype: Deployable software that integrates the technology with existing traffic management systems.*			

R&D Plan		*From a milestone perspective below is a tentative list of milestones:*
		• *Milestone 1 – Year 1*
		  ○ *Develop risk-aware multi-agent decision and safety allocation algorithms under full observability for STM/ATM scenarios.*
		  ○ *Benchmark performance in high-fidelity space and air traffic simulations (LEO conjunctions, UAM corridors) to compare architectures and select a baseline framework.*
		  ○ *Tasks: Pairwise and small-fleet conflict resolution, safety allocation under shared constraints, congestion-aware planning.*
		• *Milestone 2 – Year 2*
		  ○ *Extend the framework to partial observability, incorporating state and intent uncertainty using filtering and prediction models compatible with operational STM/ATM data feeds.*
		  ○ *Validate performance in large-scale simulated environments with heterogeneous agents and limited information sharing.*
		  ○ *Tasks: Multi-vehicle coordination under uncertainty, early*			

		maneuver decision-making, airspace-wide deconfliction.
		
		• *Milestone 3 – Year 3*
		  ○ *Deploy the decision and safety framework in hardware-in-the-loop and digital-twin testbeds representative of real STM/ATM operations, including long-horizon orbital scenarios and dense airspace traffic.*
		  ○ *Demonstrate system-level safety and efficiency gains under realistic communication and operational constraints.*
		  ○ *Tasks: Constellation-level STM coordination, integrated STM/ATM decision support, performance-constrained autonomy.*
		
		These milestones apply to both orbital and aerial traffic management use cases. While deployment in operational environments involves regulatory and integration challenges, Korea's strengths in satellite systems, UAM platforms, and systems engineering create strong opportunities for industrial partners to transition these capabilities into deployable STM/ATM services.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The Control, Coordination, and Competition under Uncertainty (C3U) Laboratory at Georgia Tech will be responsible for the development of the core decision-making, safety, and coordination algorithms that underpin the proposed STM/ATM autonomy framework. This includes the following activities:* • *Development of risk-aware multi-agent decision-making and maneuver allocation algorithms for space and air traffic management.* • *Design and analysis of formal safety certification and constraint allocation mechanisms for decentralized STM/ATM systems.* • *Development of intent-aware prediction and uncertainty-aware planning models under partial observability.* • *Integration and validation of algorithms in software, high-fidelity simulation, and digital-twin testbeds representative of orbital and airspace operations.* *The C3U Lab has an established track record in multi-agent autonomy, decision-making under uncertainty, and safety-critical systems, and currently hosts graduate researchers engaged in closely related domains. If successful, funding from this project will support collaborative research and training opportunities for Korean students and researchers, strengthening long-term Korea–GT partnerships in autonomous traffic management technologies.*
Expected R&R of Korean SME Partner	*The Korean SME satellite partner will be responsible for the integration and validation of autonomous STM technologies within realistic orbital operations environments. This includes:* • *Provision of satellite mission models, sensing configurations, and operational constraints, or representative hardware-in-the-loop platforms.* • *Integration of the distributed maneuver decision and safety allocation software with existing mission planning, flight dynamics, and ground control systems.* • *Execution of high-fidelity simulation and digital-twin evaluations for conjunction avoidance, constellation coordination, and long-horizon maneuver planning.* • *Benchmarking of safety, fuel efficiency, and operational throughput against current operator-in-the-loop practices to assess deployment readiness.* *The Korean SME aerial vehicle or UAM partner will be responsible for deploying and testing autonomous ATM decision-support technologies in dense airspace scenarios. This includes:* • *Provision of aerial vehicle platforms, sensing suites, and flight performance models, or equivalent testbed infrastructure.* • *Integration of the distributed traffic deconfliction and coordination software with onboard autonomy and ground-based traffic management systems.* • *Conduct of indoor, outdoor, and corridor-based flight testing under varying traffic densities and communication conditions.*

	• *Validation of safety, throughput, and delay reduction relative to existing air traffic management workflows.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The expected outcomes will include:* 1. *A deployable autonomy framework for STM/ATM enabling safe coordination of dense satellite and aerial traffic.* 2. *Provable safety and risk management tools that reduce reliance on human-in-the-loop decision making.* 3. *Improved operational efficiency through earlier maneuver decisions, reduced fuel use, and lower delay.* 4. *Validated software prototypes and digital-twin testbeds transferable to Korean industry partners.* 5. *Accelerated commercialization of autonomous traffic management services for LEO constellations and UAM.* 6. *Strengthened Korea–GT research collaboration and workforce development in autonomous systems and space–air mobility.*

6. References

(1) S. H. Q. Li et al., “Allocating Corrective Control to Mitigate Multi-Agent Safety Violations Under Private Preferences,” submitted to IEEE Robotics and Automation Letters, 2026. doi:10.2514/6.2026-0407.

(2) S. H. Q. Li et al., “Risk-Aware Decision-Making for Autonomous Orbital Maneuver Planning,” AIAA SciTech 2026 Forum, Orlando, FL, USA, 2026. arXiv:2512.09213.

(3) S. H. Q. Li et al., “Intent-Aware Multi-Agent Coordination Under Partial Observability,” 2026 IEEE International Conference on Robotics and Automation (ICRA), accepted, 2026. arXiv:2509.14063.

(4) S. H. Q. Li et al., “Sequential Decision-Making for Autonomous Collision Avoidance in Orbital Operations,” AAS/AIAA Astrodynamics Specialist Conference, Boston, MA, USA, 2025. arXiv:2508.05876.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization